

Continuous Assessment III

Paul Haskins

April 10, 2025

Question 1

0.0.1 (a)

Firstly we will normalize the given wave function

$$\psi(x, y) = ce^{-\frac{1}{2}\alpha r^2} \quad (1)$$

$$\langle \psi | \psi \rangle = 1 = \int |\psi|^2 dx \quad (2)$$

$$1 = \int_0^\infty \int_0^{2\pi} |c|^2 e^{-\alpha r^2} r dr d\theta \quad (3)$$

$$\text{let } u = r^2 \implies du = 2r dr, 1 = 2\pi |c|^2 \int_0^\infty e^{-\alpha u} \frac{1}{2} du \quad (4)$$

$$1 = |c|^2 \pi \left[-\frac{1}{\alpha} e^{-\alpha u} \right]_0^\infty = |c|^2 \frac{\pi}{\alpha} \quad (5)$$

$$\implies c = \pm \sqrt{\frac{\alpha}{\pi}} \quad (6)$$

$$\implies \psi(x, y) = \sqrt{\frac{\alpha}{\pi}} e^{-\frac{1}{2}\alpha r^2} \quad (7)$$

$$(8)$$

We have the Hamiltonian

$$\mathcal{H} = \frac{\hbar^2}{2m} \vec{\nabla}^2 + \frac{1}{2} m \omega^2 r^2 + br \quad (9)$$

$$(10)$$

We have the kinetic term

$$\hat{T} = \frac{\hbar^2}{2m} \vec{\nabla}^2 \quad (11)$$

$$\langle \hat{T} \rangle = \left\langle \frac{\hbar^2}{2m} \vec{\nabla}^2 \right\rangle \quad (12)$$

As our wave function only has radial dependence we only need to bother with the radial part of the gradient. We can also work with the non-normalised wavefunction as the general expression for the expected value

$$\langle \hat{A} \rangle = \frac{\langle \psi | \hat{A} | \psi \rangle}{\langle \psi | \psi \rangle} \quad (13)$$

of a wave function is would cancel out the pre-factor so using the non-normalized makes algebra nicer.

$$\left(\frac{d^2}{dr^2} + \frac{1}{r} \frac{d}{dr} \right) \psi = \frac{d}{dr} \left(-\alpha r e^{-\frac{1}{2}\alpha r^2} \right) + \frac{1}{r} \left(-\alpha r e^{-\frac{1}{2}\alpha r^2} \right) = e^{-\frac{1}{2}\alpha r^2} (-2\alpha + \alpha^2 r^2) \quad (14)$$

$$\langle \hat{T} \rangle = \left\langle \frac{\hbar^2}{2m} \vec{\nabla}^2 \right\rangle = -\frac{\hbar^2}{2m} \int d^2 r \psi_{norm} \left(\vec{\nabla}^2 \psi \right) \quad (15)$$

$$\langle \hat{T} \rangle = -\frac{\hbar^2}{2m} \langle \alpha^2 r^2 - 2\alpha \rangle = -\frac{\hbar^2}{2m} [\alpha^2 \langle r^2 \rangle - 2\alpha] \quad (16)$$

Now we want to find $\langle r^2 \rangle = \int r^2 |\psi_{norm}|^2 d^2 r$.

$$\langle r^2 \rangle = \int_0^{2\pi} d\theta \int_0^\infty r^3 \frac{\alpha}{\pi} e^{-\alpha r^2} dr \quad u = r^2 \implies \langle r^2 \rangle = 2\alpha \int_0^\infty e^{-\alpha u} u \frac{du}{2} \quad (17)$$

$$\langle r^2 \rangle = \alpha \left[-\frac{p}{\alpha} e^{-\alpha p} \right]_0^\infty - \alpha \int_0^\infty -\frac{dp}{\alpha} e^{-\alpha p} = -\frac{1}{\alpha^2} e^{-\alpha p} \Big|_0^\infty = \frac{1}{\alpha} \quad (18)$$

$$\implies \langle \hat{T} \rangle = -\frac{\hbar^2}{2m} \langle \alpha^2 r^2 - 2\alpha \rangle = -\frac{\hbar^2}{2m} [\alpha - 2\alpha] = \frac{\hbar^2 \alpha}{2m} \quad (19)$$

For the potential term we have

$$\langle \hat{V} \rangle = \left\langle \frac{1}{2} m \omega^2 r^2 + b r \right\rangle = \frac{1}{2} m \omega^2 \langle r^2 \rangle + b \langle r \rangle \quad (20)$$

Now we want to find $\langle r \rangle = \int r |\psi_{norm}|^2 d^2 r$.

We can use a handy generalised Gaussian given by

$$\int_{-\infty}^{\infty} x^{2n} e^{-\alpha x^2} dx = \sqrt{\frac{\pi}{\alpha}} \frac{(2n-1)!!}{(2\alpha)^n} \quad (21)$$

Where the double factorial is the same as a single factorial but only keeps integers of the same parity of $2n-1$.

$$\langle r \rangle = \int_0^{2\pi} d\theta \int_0^{\infty} r^2 \frac{\alpha}{\pi} e^{-\alpha r^2} dr = \alpha \int_{-\infty}^{\infty} r^2 e^{-\alpha r^2} dr \quad (22)$$

$$\langle r \rangle = \alpha \sqrt{\frac{\pi}{\alpha}} \frac{3}{2\alpha} = \sqrt{\frac{\pi}{4\alpha}} \quad (23)$$

$$\implies \langle \hat{V} \rangle = \frac{1}{2} m \omega^2 \left(\frac{1}{\alpha} \right) + b \sqrt{\frac{\pi}{4\alpha}} \quad (24)$$

$$E = \langle \mathcal{H} \rangle = \langle \hat{T} \rangle + \langle \hat{V} \rangle = \frac{1}{2} \left(\frac{\hbar^2 \alpha}{m} + m \omega^2 \alpha^{-1} + b \sqrt{\pi} \alpha^{-\frac{1}{2}} \right) \quad (25)$$

If we want to minimise wrt α

$$0 = \frac{1}{2} \left(\frac{\hbar^2}{m} - m \omega^2 \alpha^{-2} - \frac{1}{2} b \sqrt{\pi} \alpha^{-\frac{3}{2}} \right) \quad (26)$$

$$\frac{2\hbar^2}{m} \alpha^2 - b \sqrt{\pi} \alpha - 2m \omega^2 = 0 \quad (27)$$

If we set $b = 0$

$$\implies \frac{2\hbar^2}{m} \alpha^2 - 2m \omega^2 = 0 \quad (28)$$

$$\alpha^2 = \frac{m^2 \omega^2}{\hbar^2}, \text{ so } \alpha = \frac{m \omega}{\hbar} \quad (29)$$

where we take pos as $\alpha > 0$

$$E(\alpha) = \langle \mathcal{H} \rangle = \langle \hat{T} \rangle + \langle \hat{V} \rangle = \frac{1}{2} \left(\frac{\hbar^2 \alpha}{m} + m \omega^2 \alpha^{-1} + (b=0) \sqrt{\pi} \alpha^{-\frac{1}{2}} \right) \quad (30)$$

$$E(\alpha) = \langle \hat{T} \rangle + \langle \hat{V} \rangle = \frac{1}{2} \left(\frac{\hbar^2 \frac{m \omega}{\hbar}}{m} + m \omega^2 \left(\frac{m \omega}{\hbar} \right)^{-1} \right) \quad (31)$$

$$E = \frac{1}{2} (\hbar \omega + \hbar \omega) = \hbar \omega \quad (32)$$

Which we recognise as the 2D harmonic oscillator.

Question 2

0.0.2 (a)

$$\hat{H} = H_1^0 + H_1^0 + \hat{V}_{int} \quad (33)$$

$$\hat{H} = \sum_{i=1}^2 \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx_i^2} + \frac{1}{2} m \omega^2 x_i^2 \right) + \hat{V}_{int} \quad (34)$$

$$E_n = \hbar \omega \left(n_i + \frac{1}{2} \right) \quad n_i = 0, 1 \quad (35)$$

$$E_n = \hbar \omega \left[\left(n_1 + \frac{1}{2} \right) + \left(n_2 + \frac{1}{2} \right) \right] \quad n_1, n_2 = 0, 1, 2, \dots \quad (36)$$

$$\implies E_0^0 = \hbar \omega \left[\frac{1}{2} + \frac{1}{2} \right] = \hbar \omega \quad (37)$$

We know

$$E \leq E_0^0 + \langle \psi | V_{int} | \psi \rangle \quad (38)$$

We have the standard harmonic oscillator ground state wave function

$$\psi_0(x_i) = \left(\frac{m\omega}{\pi\hbar} \right)^{\frac{1}{4}} e^{-\frac{m\omega}{2\hbar} x_i^2} \quad (39)$$

we can separate the wave functions

$$\psi(x_1, x_2) = \psi_0(x_1) \psi_0(x_2) \quad (41)$$

$$(42)$$

$$\psi_0(x_i) = \left(\frac{m\omega}{\pi\hbar} \right)^{\frac{1}{2}} e^{-\frac{m\omega}{2\hbar} x_1^2} e^{-\frac{m\omega}{2\hbar} x_2^2} \quad (43)$$

$$\langle V_{int} \rangle = V_0 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dx_1 dx_2 |\psi_0(x_1)|^2 |\psi_0(x_2)|^2 \delta(|x_1 - x_2| - a) \quad (44)$$

$$\text{using } \delta(|x_1 - x_2| - a) = \delta(x_1 - x_2 - a) + \delta(x_1 - x_2 + a) \quad (45)$$

$$\langle V_{int} \rangle = V_0 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dx_1 dx_2 |\psi_0(x_1)|^2 |\psi_0(x_2)|^2 \delta(x_1 - x_2 - a) + V_0 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dx_1 dx_2 |\psi_0(x_1)|^2 |\psi_0(x_2)|^2 \delta(x_1 - x_2 + a) \quad (46)$$

$$\langle V_{int} \rangle = V_0 \int_{-\infty}^{\infty} dx_1 |\psi_0(x_1)|^2 |\psi_0(x_1 + a)|^2 + V_0 \int_{-\infty}^{\infty} dx_1 |\psi_0(x_1)|^2 |\psi_0(x_1 - a)|^2 \quad (47)$$

$$|\psi_0(x)|^2 = |\psi_0(-x)|^2 \quad (48)$$

$$\langle V_{int} \rangle = 2V_0 \int_{-\infty}^{\infty} dx_1 |\psi_0(x_1)|^2 |\psi_0(x_1 + a)|^2 \quad (49)$$

$$(50)$$

$$= 2V_0 \int_{-\infty}^{\infty} dx_1 \left(\frac{m\omega}{\pi\hbar} \right) e^{-\frac{m\omega}{\hbar} x_1^2} e^{-\frac{m\omega}{\hbar} (x_1 + a)^2} = 2V_0 \left(\frac{m\omega}{\pi\hbar} \right) \int_{-\infty}^{\infty} dx_1 e^{-\frac{m\omega}{\hbar} x_1^2} e^{-\frac{m\omega}{\hbar} (x_1^2 + a^2 + 2ax_1)} \quad (51)$$

$$= 2V_0 \left(\frac{m\omega}{\pi\hbar} \right) e^{-\frac{m\omega}{\hbar} a^2} \int_{-\infty}^{\infty} dx_1 e^{-\frac{2m\omega}{\hbar} (x_1^2 + ax_1)} = 2V_0 \left(\frac{m\omega}{\pi\hbar} \right) e^{-\frac{m\omega}{\hbar} a^2} \int_{-\infty}^{\infty} dx_1 e^{-\frac{2m\omega}{\hbar} (x_1 + \frac{a}{2})^2 + \frac{m\omega a^2}{2}} \quad (52)$$

$$= 2V_0 \left(\frac{m\omega}{\pi\hbar} \right) e^{-\frac{m\omega}{2\hbar} a^2} \int_{-\infty}^{\infty} dx_1 e^{-\frac{2m\omega}{\hbar} (x_1 + \frac{a}{2})^2} = 2V_0 \left(\frac{m\omega}{\pi\hbar} \right) e^{-\frac{m\omega}{2\hbar} a^2} \sqrt{\frac{\pi\hbar}{2m\omega}} \quad (53)$$

$$\implies \langle V_{int} \rangle = \sqrt{\frac{2V_0^2 m\omega}{\pi\hbar}} e^{-\frac{m\omega}{2\hbar} a^2} \quad (54)$$

$$(55)$$

So we have an upper bound on the energy given by

$$E_{gs} \leq \hbar\omega + \sqrt{\frac{2V_0^2 m\omega}{\pi\hbar}} e^{-\frac{m\omega}{2\hbar} a^2} \quad (56)$$

$$(57)$$

Question 3

0.0.3 (a)

We know that for a particle in a 1D infinite potential well we have

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) \quad (58)$$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}, \quad n = 1, 2, \dots \quad (59)$$

Given our potential we have the perturbed Hamiltonian

$$\mathcal{H}' = -\epsilon(2x - a), \text{ for } 0 \leq t \leq \tau \quad (60)$$

We have the transition from the ground state $|1\rangle$ to the first excited state $|2\rangle$ given by

$$c_{12} = \frac{i}{\hbar} \int_0^\tau \langle 2 | \mathcal{H}' | 1 \rangle e^{i\omega_{21}t} dt = \frac{i}{\hbar} \int_0^a \psi_2^* [-\epsilon(2x - a)] \psi_1 e^{i\omega_{10}t} dx \quad (61)$$

$$c_{12} = \frac{i}{\hbar} \frac{2}{a} \int_0^a \sin\left(\frac{2\pi x}{a}\right) \sin\left(\frac{\pi x}{a}\right) e^{i\omega_{10}t} [-\epsilon(2x - a)] dx \quad (62)$$

We can use

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)] \quad (63)$$

XXX before

$$V_{12} = \frac{1}{a} \int_0^a [-\epsilon(2x - a)] \left[\cos\left(\frac{\pi x}{a}\right) - \cos\left(\frac{3\pi x}{a}\right) \right] dx = \quad (64)$$

$$I_n = \int_0^a (2x - a) \cos\left(\frac{n\pi x}{a}\right) dx = \left[\frac{a}{n\pi} \sin\left(\frac{n\pi x}{a}\right) (2x - a) \right] \Big|_0^a - \int_0^a \frac{a}{n\pi} 2dx \sin\left(\frac{n\pi x}{a}\right) \quad (65)$$

$$I_n = \frac{a^2}{n\pi} \sin(n\pi) + \left[\frac{2a^2}{n^2\pi^2} \cos\left(\frac{n\pi x}{a}\right) \right] \Big|_0^a = \frac{a^2}{n\pi} \sin(n\pi) + \frac{2a^2}{n^2\pi^2} [(-1)^n - 1] \quad (66)$$

$$I_1 = -\frac{4a^2}{\pi^2}, \text{ and } I_3 = -\frac{4a^2}{9\pi^2} \quad (67)$$

$$V_{12} = -\frac{\epsilon}{a} \left[-\frac{4a^2}{\pi^2} + \frac{4a^2}{9\pi^2} \right] = \frac{32a\epsilon}{9\pi^2} \quad (68)$$

We have the coefficient given by

$$c_{12} = -\frac{i}{\hbar} \int_0^\tau dt e^{i\omega_{12}t} V_{12} \quad (69)$$

$$\omega_{12} = \frac{E_2 - E_1}{\hbar} = \frac{3\pi^2\hbar}{2ma^2} \quad (70)$$

$$c_{12} = -\frac{i}{\hbar} V_{12} \left[\frac{1}{i\omega_{12}} e^{i\omega_{12}t} \right] \Big|_0^\tau = -\frac{1}{\hbar} \left(\frac{32a\epsilon}{9\pi^2} \right) \left[\frac{2ma^2}{3\pi^2\hbar} \left(e^{i\omega_{12}\tau} - 1 \right) \right] 2 \quad (71)$$

$$|c_{12}|^2 = \left(\frac{64ma^3\epsilon}{27\hbar\pi^4} \right)^2 (\cos(\omega_{12}\tau) - 1)^2 + \sin^2(\omega_{12}\tau) \quad (72)$$

$$|c_{12}|^2 = \left(\frac{-64ma^3\epsilon}{27\hbar\pi^4} \right)^2 (2 - 2\cos(\omega_{21}\tau)) \quad (73)$$

$$= \left(\frac{-64ma^3\epsilon}{27\hbar\pi^4} \right)^2 4\sin^2\left(\frac{\omega_{12}\tau}{2}\right) \text{ So the probability is given by} \quad (74)$$

$$\mathcal{P}_{12} = \frac{16384m^2a^6\epsilon^2}{729\hbar^4\pi^8} \sin^2\left(\frac{3}{4} \frac{\hbar^2\pi^2}{ma^2} \tau\right) \quad (75)$$